Technical proposal: FaceAttend EDU

Jonattan Steven Rizo Solano

Diego Andrés Gutiérrez Núñez

Juan David Arboleda Perdomo

Centro de la Industria, la Empresa y los Servicios, Servicio Nacional de Aprendizaje

Technologist in Software Analysis and Development, Cohort 3145556

Karol Daniela Correa

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Contents

1. EXECUTIVE SUMMARY 6
1.1 Project Description 6
1.2 General Objective 6
1.3 Specific Objectives 6
1.4 Scope 7
1.5 Personnel Involved 8
1.6 Summary 8
2. INTRODUCTION 8
2.1 Problem Statement 8
2.2 Purpose 9
2.3 Justification 9
3. PROJECT DESCRIPTION 10
3.1 General Description 10
3.2 Product Perspective 10
3.3 User Characteristics 10
3.4 Main Features 11
4. REQUIREMENTS 11
4.1 Software Requirements 11
4.2 Hardware Requirements 15
5. PROPOSED SOLUTION 16
5.1 Solution Overview 16
5.2 Solution Pillars 16
5.3 Product Roadmap 17
5.4 Design Principles 17
6. SYSTEM ARCHITECTURE 17
6.1 Architectural Style 17

6.2 Container Diagram (C4) 17
6.3 Microservices Catalog 18
6.4 Hexagonal Architecture 19
6.5 Adopted Architectural Patterns 20
6.6 Docker Networks 20
7. DATA MODEL 21
7.1 Summary 21
7.2 PostgreSQL Schemas 21
7.3 MongoDB Collections 21
7.4 Main Entities by Context 22
7.5 Design Rules 24
7.6 Business Rules (Summary) 24
8. API AND MICROSERVICES 25
8.1 API Gateway (Kong) 25
8.2 Detailed Biometric Endpoints 26
8.3 Authentication 26
8.4 Domain Events 27
8.5 OpenAPI Contracts 28
9. SECURITY 28
9.1 Authentication and Authorization 28
9.2 Data Protection 29
9.3 Biometric Validation 29
9.4 Security Technical Debt 29
10. INFRASTRUCTURE AND DEVOPS 30
10.1 Environments 30
10.2 Ports 30
10.3 Main Environment Variables 31

10.4 CI/CD 31
10.5 Observability 31
11. WORK PLAN 32
11.1 Methodology 32
11.2 Project Phases 32
11.3 Gantt Chart 32
11.4 Deliverables per Sprint 33
12. WORK TEAM 33
12.1 Members 33
12.2 Instructor 34
13. BUDGET 34
13.1 Human Resources 34
13.2 Vendors and Technologies 34
13.3 Budget Summary 35
14. QUALITY MANAGEMENT 35
14.1 Applied Standards 35
14.2 Testing Strategy 36
14.3 Non-Functional Requirements 36
15. RISK MANAGEMENT 37
15.1 Identified Risks 37
15.2 Registered Technical Debt 37
16. ANNEXES 38
16.1 Glossary of Terms 38
16.2 References 39
16.3 Repository 39
Activity plan 40
Roles (Scrum) 40

Activity Plan 40
Hardware requirements 42
Sheet: Developers 42
Sheet: Software 49
Sheet: Users 54

Technical proposal: FaceAttend EDU

1. EXECUTIVE SUMMARY

1.1 Project Description

FaceAttend EDU is a comprehensive attendance management platform using dual biometric recognition: facial recognition using the device camera (computer or mobile) and fingerprint recognition using the DigitalPersona 4500 SDK. Designed specifically for educational institutions, the system automates student check-in and check-out, centralizes academic histories, supports justification flows, and generates role-based reports with automatic alerts.

1.2 General Objective

Develop a comprehensive software system that uses the device camera for facial recognition and the DigitalPersona 4500 SDK for fingerprint biometrics, mobile and web applications, and AI tools implemented directly in code, to efficiently validate student attendance and effectively track their participation in classes, minimizing inconsistencies in records, improving academic performance, and facilitating instructor control.

1.3 Specific Objectives

1. Design and implement an attendance registration module that integrates dual biometric technologies: facial recognition using the device camera and fingerprint recognition using the DigitalPersona 4500 SDK, implementing directly in code all necessary functionality for real-time data capture, ensuring precise and automated validation in in-person, hybrid, or virtual environments.
2. Develop an intuitive user interface that allows instructors and administrators to view attendance records, monitor class participation through interaction tools, and receive automatic alerts about absences or low participation.
3. Integrate an automated notification system via email or mobile apps that informs instructors, parents, and students about attendance or participation anomalies, facilitating immediate corrective actions and promoting accountability.

4. Ensure software scalability and compatibility with various market technologies, allowing adaptation to different educational institutions and learning environments.
5. Implement security and privacy measures in the software, including data encryption and compliance with regulations such as Law 1581 on Personal Data Protection (Colombia), to protect sensitive student information and ensure ethical use of integrated technologies (Congress of Colombia, 2012).

1.4 Scope

In scope (MVP - Horizon 1, 0-3 months):

- User management with CSV and manual registration
- Login and authentication with opaque sessionId
- Password recovery and change
- Management of environments, cohorts, courses, and academic periods
- Check-in/check-out records FACIAL + MANUAL/IMPORT
- System and academic configuration
- Justification flow with supporting documents
- Histories and reports by date range/cohort
- Facial recognition with liveness detection
- Responsive frontend (Web + Mobile)

Out of scope:

- Teacher payroll, grades, payments
- Complete physical access control (doors/turnstiles)
- Integration with external systems (enrollments/grades/IdP)
- Advanced messaging (SMS/WhatsApp)
- Predictive dropout analytics
- IoT device failure alerts
- Separate native mobile apps (MVP is responsive web + mobile web)

1.5 Personnel Involved

Table 1

1.5 Personnel Involved

Name	Role	Responsibility
Diego Andrés Gutiérrez Nuñez	Administrator, Developer, AI	Information analysis, design and programming
Juan David Arboleda Perdomo	Analyst, Database	Information analysis, design and programming
Jonattan Steven Rizo Solano	Hardware, Developer	Information analysis, design and programming

Source. Own elaboration.

1.6 Summary

The FaceAttend EDU project represents a complete technological solution that transforms attendance management in educational institutions through facial recognition, polyglot microservices architecture, and a unified frontend with React Native/Expo. The system is designed with professional quality standards, including IEEE 829, ISO 25010, ISO 29110, and ISTQB documentation (Institute of Electrical and Electronics Engineers [IEEE], 2008; International Organization for Standardization & International Electrotechnical Commission [ISO & IEC], 2011; International Organization for Standardization & International Electrotechnical Commission [ISO & IEC], 2016; International Software Testing Qualifications Board, 2018).

2. INTRODUCTION

2.1 Problem Statement

In the current educational environment, manual attendance registration presents multiple issues:

- *Inconsistencies in records:* Traditional methods (paper lists, roll calls) are prone to human error, forgetfulness, and manipulation.
- *Administrative overload:* Instructors spend valuable class time on administrative tasks instead of focusing on teaching.

- *Impersonation risk:* Without automated identity verification, it is easy for a student to register attendance for another.
- *Lack of traceability:* No centralized system exists to efficiently consult attendance histories.
- *Difficulty in report generation:* Manual data consolidation for reports is slow and error-prone.

2.2 Purpose

FaceAttend EDU emerges as a response to these issues, providing a platform that:

- Automates attendance registration through facial and biometric recognition
- Guarantees student identity through biometric verification with liveness detection
- Centralizes all attendance information in a structured database
- Generates automatic reports and real-time alerts
- Reduces instructor administrative burden
- Complies with data protection regulations (Law 1581)

2.3 Justification

The implementation of FaceAttend EDU is justified by:

1. *Operational efficiency:* Reduction of attendance registration time from minutes to seconds per student.
2. *Precision and security:* Elimination of human errors and identity impersonation through biometrics.
3. *Complete traceability:* Complete and auditable attendance histories.
4. *Academic improvement:* Early identification of absenteeism patterns affecting performance.
5. *Scalability:* Microservices architecture that allows growth with the institution.
6. *Regulatory compliance:* Alignment with Law 1581 on Personal Data Protection.

3. PROJECT DESCRIPTION

3.1 General Description

FaceAttend EDU is a web and mobile application designed for educational institutions seeking to optimize the attendance registration process through dual biometric technology: facial recognition using the device camera and fingerprint recognition using the DigitalPersona 4500 SDK. This system automates student presence verification and allows generating precise reports, improving academic management and reducing administrative burden.

The application integrates as a support tool for educational management, with functionalities adaptable to different academic levels and focused on usability, security, and efficiency. It is an autonomous solution, although it can be complemented with other institutional academic information systems.

3.2 Product Perspective

FaceAttend EDU is a system composed of:

- *Web and Mobile Frontend:* Responsive application built with React Native and Expo, compatible with modern browsers and iOS/Android mobile devices.
- *Backend:* 9 microservices specialized in different business domains.
- *API Gateway:* Kong OSS as the single entry point for all requests.
- *Databases:* PostgreSQL 17 (relational data) and MongoDB 7 (biometric data).
- *Messaging:* Apache Kafka for asynchronous communication between services.
- *Infrastructure:* Docker Compose for container orchestration.

3.3 User Characteristics

Table 2

3.3 User Characteristics

User Type	Description	Main Needs
Learner/Student	Main user who registers attendance	Quick attendance registration, history consultation, absence justification
Instructor/Teacher	Responsible for managing attendance in their classes	Automatic attendance capture, report visualization, justification management

User Type	Description	Main Needs
Administrator	Manages system configuration	User management, academic configuration, global report generation
Supervisor	Monitors academic compliance	Absenteeism alerts, consolidated reports, participation metrics

Source. Own elaboration.

3.4 Main Features

1. *Facial Attendance Registration:* Face capture with device camera (computer or mobile), liveness detection, and automatic registration with date/time.
2. *Fingerprint Attendance Registration:* Fingerprint capture via DigitalPersona 4500 SDK, in-code processing, and automatic registration with date/time.
3. *Manual/Import Registration:* Backup method for special cases or technical failures.
4. *User Management:* Complete CRUD of learners, instructors, and administrators.
5. *Academic Management:* Administration of campuses, programs, cohorts, courses, and periods.
6. *Justifications:* Complete flow of request, review, and approval/rejection of justifications.
7. *Reports and Dashboard:* Metrics visualization, Excel export, filters by date/cohort.
8. *Notifications and Alerts:* Automatic alert system for absences or anomalies.
9. *Configuration:* Academic, security, and biometric parameters.

4. REQUIREMENTS

4.1 Software Requirements

4.1.1 Operating Systems

Server:

- *Linux:* Ubuntu Server 22.04 LTS, CentOS 8 or Debian 12 (recommended for production)
- *Windows:* Windows Server 2019/2022 (alternative)
- *Containers:* Docker 24+ and Docker Compose 2+

Client (Web Browser):

- Chrome 90+
- Firefox 88+
- Edge 90+
- Safari 14+
- Brave / Opera (any recent version)

Mobile:

- Android 8.0 (API 26) or higher
- iOS 14.0 or higher

4.1.2 Backend Technologies

Table 3

4.1.2 Backend Technologies

Technology	Version	Use
Java	21 LTS	4 main microservices
Spring Boot	4.1.2	Main Java framework
Spring Security	7.x (managed by Spring Boot)	Authentication and authorization
Spring Data JPA	4.x	Java persistence
TypeScript	5.x (Node 24)	4 microservices
Fastify	5.x	TypeScript HTTP framework
pg (node-postgres)	8.x	PostgreSQL driver for the TypeScript services
Python	3.12	Biometric microservice
FastAPI	0.162	Python framework

Source. Own elaboration.

4.1.3 Frontend Technologies

Table 4

4.1.3 Frontend Technologies

Technology	Version	Use
Expo SDK	57	React Native framework

Technology	Version	Use
React	19.2.3	UI library
React Native	0.86.3	Mobile framework
React Native Web	0.21	Web support
React Navigation	7.x	Navigation
@digitalpersona/fingerprint	1.0.0	DigitalPersona fingerprint reader
@digitalpersona/websdk	1.1.0	DigitalPersona Web SDK
axios	1.16.1	HTTP client
i18next	26.0.4	Internationalization
xlsx	0.18.5	Excel export

Source. Own elaboration.

4.1.4 Databases

Table 5

4.1.4 Databases

Engine	Version	Use
PostgreSQL	17	Main database (8 schemas)
MongoDB	7	Biometric vector storage
Redis	7	Cache and rate-limiting

Source. Own elaboration.

4.1.5 Infrastructure and Tools

Table 6

4.1.5 Infrastructure and Tools

Technology	Version	Use
Docker	24+	Containers
Docker Compose	2+	Orchestration
Kong	3.6	API Gateway
Apache Kafka	3.8 (KRaft)	Event messaging
Liquibase	4.29.0	Database migrations
OpenTelemetry	1.24	Observability

Technology	Version	Use
Jenkins	2.x	CI/CD

Source. Own elaboration.

4.1.6 Biometric Technology

Table 7

4.1.6 Biometric Technology

Component	Technology	Description
Facial Recognition	Device camera + face-api.js	Facial detection and recognition using computer or mobile camera
Fingerprint SDK	DigitalPersona 4500 SDK	Fingerprint capture and processing
Storage	MongoDB 7	Facial embeddings and fingerprint templates with index
Facial Matching	Cosine similarity	128-d vector comparison
Fingerprint Matching	DigitalPersona SDK	Fingerprint template comparison

Source. Own elaboration.

Note: The system uses two complementary biometric methods: facial recognition using the device camera (computer or mobile) and fingerprint recognition using the DigitalPersona 4500 SDK. All functionality will be implemented directly in code.

4.1.7 Security and Validation

Table 8

4.1.7 Security and Validation

Mechanism	Details
Authentication	Opaque SessionId (UUID) - never JWT
Password Hashing	BCrypt with cost factor 12
Authorization	RBAC (Role-Based Access Control)
Encryption	SSL/TLS for communications
Rate Limiting	Kong (5173, 8090, 3000 origins)
Biometric validation	Liveness detection (facial) + Fingerprint capture (DigitalPersona 4500)
Data protection	Law 1581 compliance

Source. Own elaboration.

4.2 Hardware Requirements

4.2.1 Server

Table 9

4.2.1 Server

Component	Minimum	Recommended
Processor	Intel i5 (8th gen) or AMD equivalent	Intel i7 (12th gen) or AMD Ryzen 7
RAM	8 GB	16 GB
Storage	SSD 256 GB	SSD 512 GB
GPU	Not required	Not required
Network	Ethernet 100 Mbps	Ethernet 1 Gbps

Source. Own elaboration.

4.2.2 Biometric Devices

Table 10

4.2.2 Biometric Devices

Type	Specification	Use
Device camera	Webcam 720p+ (computer) or front camera (mobile)	Facial recognition
DigitalPersona 4500	USB fingerprint reader with SDK	Fingerprint recognition

Source. Own elaboration.

Note: The system uses the device camera (computer or mobile) for facial recognition and DigitalPersona 4500 for fingerprint recognition. Both biometric methods work complementarily.

4.2.3 Mobile Devices

Table 11

4.2.3 Mobile Devices

Platform	Requirements
Android	Android 8.0+, front camera 720p+, 2 GB RAM

Platform	Requirements
iOS	iOS 14.0+, front camera 720p+, 2 GB RAM

Source. Own elaboration.

4.2.4 IoT and Sensors (Future - Horizon 3)

Table 12

4.2.4 IoT and Sensors (Future - Horizon 3)

Device	Function
Motion Sensors	Verify physical presence in the classroom
NFC/Proximity Sensors	Confirm student location
Raspberry Pi	Edge processing for IoT

Source. Own elaboration.

5. PROPOSED SOLUTION

5.1 Solution Overview

The proposed solution is a comprehensive software system based on a polyglot microservices architecture that uses market-available technologies to efficiently validate student attendance and effectively track their class participation.

5.2 Solution Pillars

1. *Automated Attendance Capture:* Dual biometric registration: facial (device camera) and fingerprint (DigitalPersona 4500 SDK).
2. *Reliable Academic Tracking:* Complete histories, automatic reports, and alerts.
3. *Transparent Justification Flow:* Complete process of request, review, and resolution.
4. *Security, Privacy, and Usability:* Regulatory compliance, robust authentication, and intuitive UX.
5. *Scalable and Compatible Operation:* Microservices architecture, Docker containers, and API Gateway.

5.3 Product Roadmap

Table 13

5.3 Product Roadmap

Horizon	Period	Deliverables
Horizon 1 (MVP)	0-3 months	Identity, Academic, Scheduling, Attendance core, Biometric facial enroll (camera) + fingerprint enroll (DigitalPersona 4500), Justification workflow, Notification in-app
Horizon 2 (Traceability)	3-6 months	End-to-end justifications, dashboards, alerts, per-school configuration
Horizon 3 (Scale)	6-12 months	Advanced analytics, multi-school tenancy, SDK optimization

Source. Own elaboration.

5.4 Design Principles

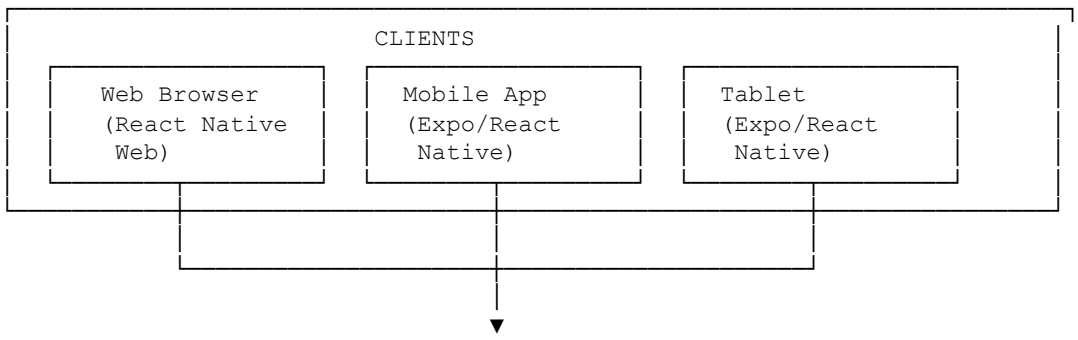
- *Precision over speed:* Recognition accuracy is paramount.
- *Privacy by design:* Biometric data treated as sensitive PII.
- *Offline-first (future):* Offline capture capability with later synchronization.
- *API-First:* OpenAPI contracts as source of truth.
- *Observability by Design:* Logs, metrics, and tracing from day one.

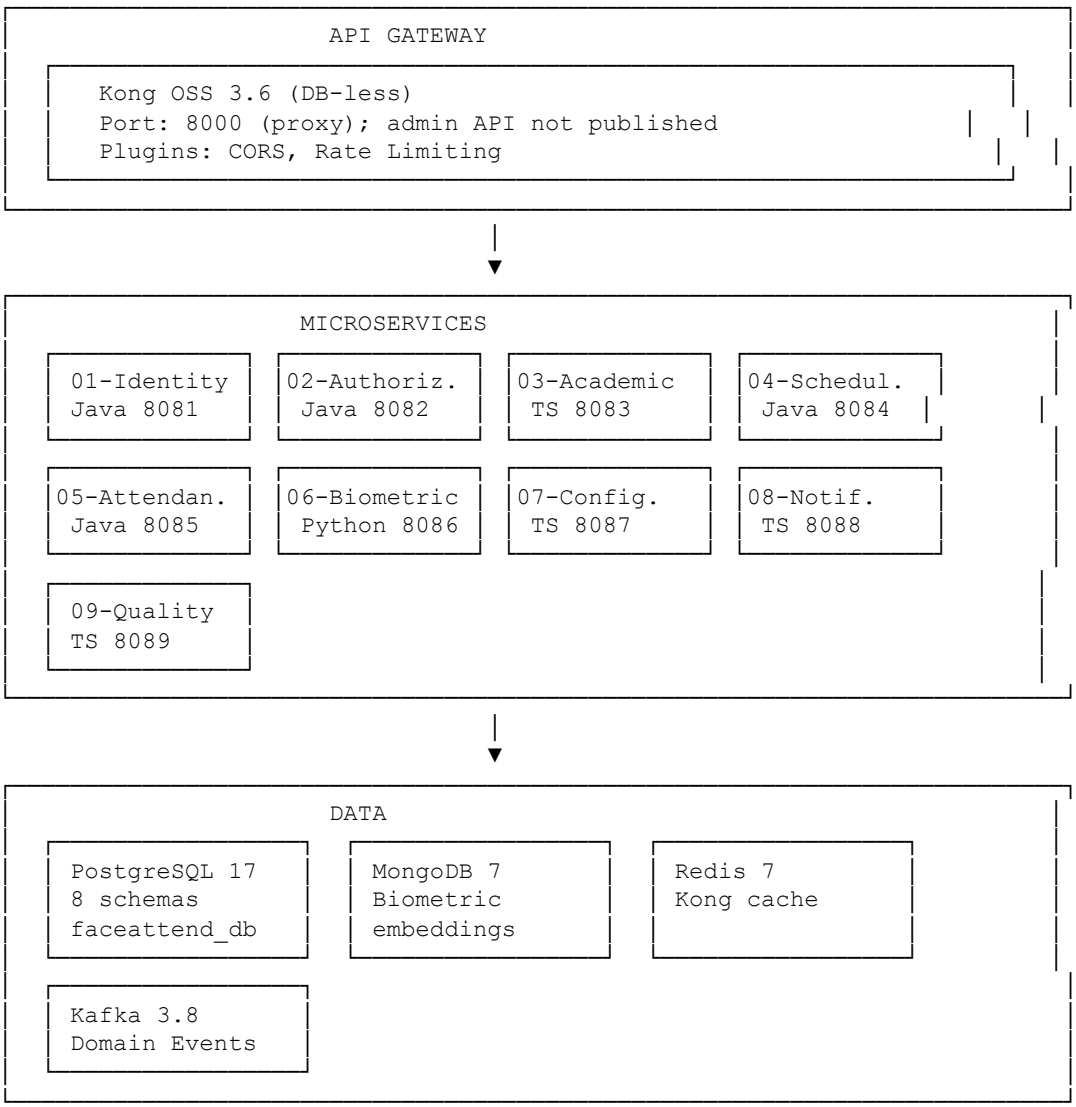
6. SYSTEM ARCHITECTURE

6.1 Architectural Style

The system implements a *Polyglot Microservices* architecture with *Hexagonal Architecture* (Ports and Adapters) and *Domain-Driven Design* (Bounded Contexts).

6.2 Container Diagram (C4)





6.3 Microservices Catalog

Table 14

6.3 Microservices Catalog

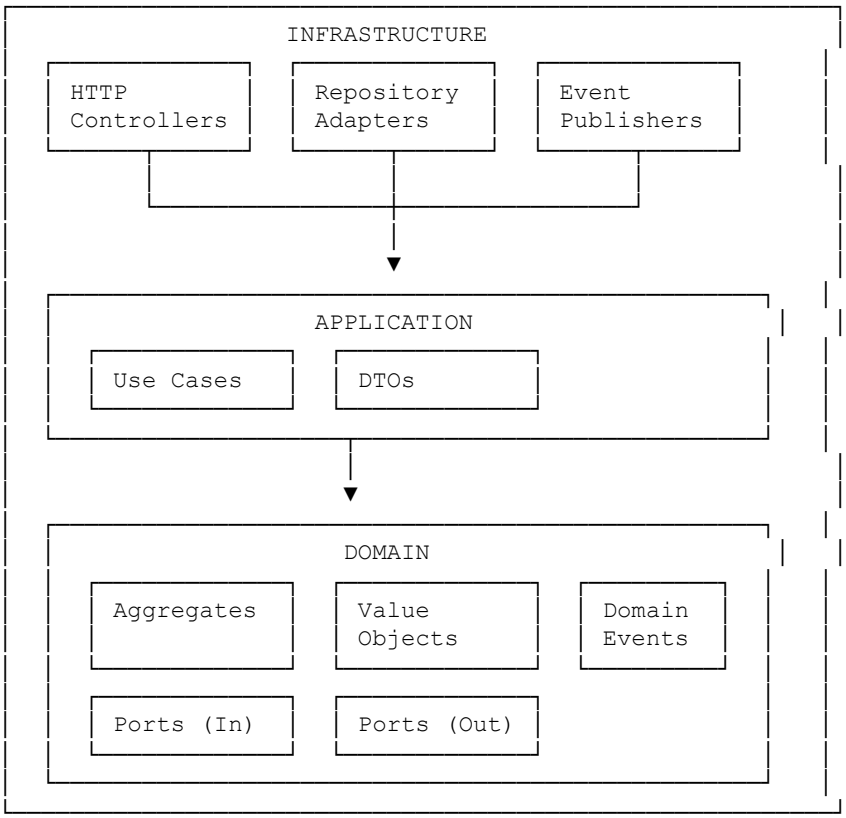
#	Service	Port	Language	Framework	Database	Responsibility
01	ms-identity	8081	Java 21	Spring Boot 4.1	PostgreSQL	Persons, users, sessions, password policies
02	ms-authorization	8082	Java 21	Spring Boot 4.1	PostgreSQL	RBAC (roles, permissions, assignments)
03	ms-academic	8083	TypeScript	Fastify + pg	PostgreSQL	Campuses, programs, cohorts, courses, enrollments

#	Service	Port	Language	Framework	Database	Responsibility
04	ms-scheduling	8084	Java 21	Spring Boot 4.1	PostgreSQL	Environments, schedule blocks, class sessions
05	ms-attendance	8085	Java 21	Spring Boot 4.1	PostgreSQL	Attendance records, justifications, documents
06	ms-biometric	8086	Python 3.12	FastAPI	MongoDB	Facial embeddings + fingerprint templates, matching
07	ms-configuration	8087	TypeScript	Fastify	PostgreSQL	Academic/security configuration, biometric cases
08	ms-notification	8088	TypeScript	Fastify	PostgreSQL	Alerts and notifications
09	ms-quality	8089	TypeScript	Fastify	PostgreSQL	ISO 25010/29110 quality evaluation, ISTQB
99	api-gateway	8000	-	Kong OSS 3.6	Redis	Routing, auth, rate-limit, CORS

Source. Own elaboration.

6.4 Hexagonal Architecture

Each microservice follows the Ports and Adapters pattern:



Dependency rule: `infrastructure → application → domain` (always inward).

6.5 Adopted Architectural Patterns

Table 15

6.5 Adopted Architectural Patterns

Pattern	Status	Description
API Gateway (Kong OSS)	Implemented	Single entry point with CORS and rate-limiting
Database per Service (schema-per-context)	Implemented	Each bounded context has its own schema in PostgreSQL
Hexagonal Architecture	Implemented	Ports and adapters in all services
Message Broker (Kafka)	Implemented	Asynchronous communication between services
Circuit Breaker	Partial	Defined but not fully implemented
Saga (choreographed)	Partial	For distributed operations
Outbox Pattern	Partial	For reliable event publishing
Retry + Exponential Backoff + DLQ	Implemented	For communication resilience
CQRS	Not implemented	Future
Event Sourcing	Not implemented	Future

Source. Own elaboration.

6.6 Docker Networks

Table 16

6.6 Docker Networks

Network	Members	Purpose
`faceattend-edge`	frontend-web, kong-gateway	External contact point
`faceattend-app`	kong-gateway, 9 ms-*, redis	Inter-service communication
`faceattend-data`	9 ms-*, postgres, mongodb, kafka, liquibase	Data access

Source. Own elaboration.

7. DATA MODEL

7.1 Summary

- *29 SQL tables* in PostgreSQL 17 (8 schemas)
- *2 MongoDB collections* for biometric data (facial + fingerprint)
- *28 entities + 4 Value Objects*
- *Normalization:* 3NF in the core with 0 intentional denormalizations

7.2 PostgreSQL Schemas

Table 17

7.2 PostgreSQL Schemas

Schema	Tables	Description
identity	person, app_user, user_session, password_policy	Identity and credentials
authorization	role, permission, role_permission, user_role	RBAC
academic	school, program, academic_period, cohort, course, academic_actor_type, academic_actor, enrollment	Academic structure
scheduling	environment, schedule_block, class_session	Schedules and sessions
attendance	attendance_record, justification_type, justification, supporting_document, attendance_report	Attendance
configuration	academic_configuration, security_configuration, biometric_update_case	Configuration
notification	alert_type, alert	Alerts
biometric	(empty - uses MongoDB)	Schema only
quality	quality_project, quality_evaluation, quality_evaluation_item, process_assessment, process_assessment_rating, istqb_assessment, istqb_assessment_item	Quality

Source. Own elaboration.

7.3 MongoDB Collections

Table 18

7.3 MongoDB Collections

Collection	Key Fields
facial_embedding	person_id, template_version, encoding (128-d), model_version, enrolled_at, is_active
fingerprint_embedding	person_id, finger_number, template_version, encoding, model_version, enrolled_at, is_active

Source. Own elaboration.

7.4 Main Entities by Context

Identity

- *Person:* UUID, document_type, document_number (unique), primer_nombre, segundo_nombre, primer_apellido, segundo_apellido, email, phone, birth_date, status
- *App_User:* UUID, username, password_hash, status, last_login_at, failed_attempts, locked_until
- *User_Session:* UUID, session_id (opaque), user_id, status, start_date, end_date, source_ip, user_agent
- *Password_Policy:* UUID, min_length, require_uppercase, require_lowercase, require_digit, require_special, max_age_days, history_count

Authorization

- *Role:* UUID, name, description, status
- *Permission:* UUID, name (format: context.resource:action), description
- *Role_Permission:* role_id, permission_id (composite PK)
- *User_Role:* user_id, role_id (composite PK)

Academic

- *School:* UUID, name, code, address, city_id, status
- *Program:* UUID, school_id, name, code, duration_hours, status
- *Academic_Period:* UUID, school_id, name, start_date, end_date, status
- *Cohort:* UUID, program_id, instructor_id, name, code, start_date, end_date, status
- *Course:* UUID, school_id, name, code, credits, status

- *Academic_Actor_Type:* UUID, name (INSTRUCTOR, LEARNER, etc.)
- *Academic_Actor:* UUID, person_id, actor_type_id, status
- *Enrollment:* UUID, cohort_id, learner_id, status, enrollment_date

Scheduling

- *Environment:* UUID, school_id, name, capacity, status
- *Schedule_Block:* UUID, environment_id, day_of_week, start_time, end_time
- *Class_Session:* UUID, cohort_id, course_id, environment_id, session_date, status (Open/Closed/Cancelled)

Attendance

- *Attendance_Record:* UUID, class_session_id, learner_id, status (Present/Absent/Late/Justified), recording_method (FACIAL/MANUAL/IOT/IMPORT), recorded_at, confidence_score
- *Justification_Type:* UUID, name, description, max_days, requires_document
- *Justification:* UUID, attendance_record_id, justification_type_id, reason, status (Pending/Approved/Rejected), reviewed_by, reviewed_at
- *Supporting_Document:* UUID, justification_id, file_name, file_path, file_size, mime_type, uploaded_at
- *Attendance_Report:* UUID, cohort_id, generated_at, filters (JSONB), result (JSONB)

Biometric

- *Facial_Embedding (MongoDB):* person_id, template_version, encoding (128-d array), model_version, enrolled_at, is_active
- *Fingerprint_Embedding (MongoDB):* person_id, finger_number, template_version, encoding, model_version, enrolled_at, is_active
- *Biometric_Update_Case (SQL):* UUID, person_id, update_type, status (Pending/In_Review/Approved/Rejected), requested_by, reviewed_by

Notification

- *Alert_Type:* UUID, name, severity (INFO/WARNING/CRITICAL), channel (DASHBOARD/EMAIL/PUSH)
- *Alert:* UUID, alert_type_id, user_id, title, message, status, created_at, resolved_at

Configuration

- *Academic_Configuration:* UUID, school_id, attendance_threshold, late_tolerance_minutes, justification_window_days
- *Security_Configuration:* UUID, session_timeout_minutes, max_failed_attempts, lockout_duration_minutes, password_policy_id

7.5 Design Rules

- *No FK between contexts* (only comments, references by UUID)
- *UUID* for cross-context entities
- *Soft delete* with `deleted_at`
- *Audit:* created_at, updated_at, deleted_at, created_by, updated_by, deleted_by, row_version
- *ENUMs* native to PostgreSQL
- *Naming:* snake_case, singular tables

7.6 Business Rules (Summary)

Table 19

7.6 Business Rules (Summary)

ID	Rule
RN-01	App_User must have an associated Person
RN-07	Every App_User must have at least one role
RN-14	Cohort must have a valid instructor before registering attendance
RN-18	Schedule_Block cannot overlap in the same environment
RN-22	One attendance_record per actor per session
RN-23	Attendance is only registered in sessions with Open status
RN-28	Only one facial embedding and one active fingerprint template per person and finger

ID	Rule
RN-32	Justification is 1:1 with attendance_record
RN-36	Approved justification → attendance_status = Justified

Source. Own elaboration.

8. API AND MICROSERVICES

8.1 API Gateway (Kong)

Table 20

8.1 API Gateway (Kong)

Service	Routes	Description
identity-service	`/api/v1/persons`, `/api/v1/users`, `/api/v1/auth`, `/api/v1/sessions`, `/api/v1/password-policies`	Identity and authentication
authorization-service	`/api/v1/roles`, `/api/v1/permissions`, `/api/v1/user-roles`	RBAC
academic-service	`/api/v1/schools`, `/api/v1/programs`, `/api/v1/academic-periods`, `/api/v1/cohorts`, `/api/v1/courses`, `/api/v1/actor-types`, `/api/v1/academic-actors`, `/api/v1/enrollments`	Academic structure
scheduling-service	`/api/v1/environments`, `/api/v1/schedule-blocks`, `/api/v1/class-sessions`	Schedules
attendance-service	`/api/v1/attendance-records`, `/api/v1/attendance-reports`, `/api/v1/justifications`, `/api/v1/justification-types`, `/api/v1/supporting-documents`	Attendance
biometric-service	`/api/v1/biometric`	Dual biometrics: facial (camera) + fingerprint (DigitalPersona 4500)
configuration-service	`/api/v1/academic-configurations`, `/api/v1/security-configurations`, `/api/v1/biometric-update-cases`	Configuration
notification-service	`/api/v1/alert-types`, `/api/v1/alerts`	Alerts
quality-service	`/api/v1/quality`	Quality

Source. Own elaboration.

8.2 Detailed Biometric Endpoints

Table 21

8.2 Detailed Biometric Endpoints

Method	Endpoint	Description
POST	`/api/v1/biometric/facial/enroll`	Register face (device camera)
POST	`/api/v1/biometric/facial/verify`	Verify facial identity (1:1)
POST	`/api/v1/biometric/facial/identify`	Identify facial person (1:N)
GET	`/api/v1/biometric/facial/{person_id}`	Get active facial template
GET	`/api/v1/biometric/facial/{person_id}/history`	Facial template history
DELETE	`/api/v1/biometric/facial/{person_id}`	Delete facial template
POST	`/api/v1/biometric/fingerprint/enroll`	Register fingerprint (DigitalPersona 4500)
POST	`/api/v1/biometric/fingerprint/verify`	Verify fingerprint identity (1:1)
POST	`/api/v1/biometric/fingerprint/identify`	Identify fingerprint person (1:N)
GET	`/api/v1/biometric/fingerprint/{person_id}`	Get active fingerprint template
GET	`/api/v1/biometric/fingerprint/{person_id}/history`	Fingerprint template history
DELETE	`/api/v1/biometric/fingerprint/{person_id}`	Delete fingerprint template
WS	`/api/v1/biometric/ws`	WebSocket channel

Source. Own elaboration.

8.3 Authentication

Mechanism: `Authorization: Bearer <sessionId>` where sessionId is an opaque UUID (never JWT).

Login Flow:

1. `POST /api/v1/auth/login` with `{identifier (email|username), password}`
2. Server validates credentials with BCrypt
3. A `user_session` is created with UUID sessionId
4. Response: `{sessionId, userId, sessionStatus, startDate, endDate, sourceIp}`

5. Failures always return 401 "Invalid email or password" (no user enumeration)

Session lifecycle:

- 480-minute lifetime (`session_timeout_minutes`) from `start_date`; expiry is lazy: it is applied when the session is read.
- `POST /api/v1/auth/refresh` with `{sessionId}` rotates the session: it returns a new sessionId and closes the old one (single use; 401 if expired or already refreshed).
- `POST /api/v1/auth/logout?sessionId=<UUID>` closes the session (204).

Validation:

- Kong (edge): a global Lua *pre-function* requires `Bearer <UUID>` on every non-public route (public: login, login/verify, refresh, logout, forgot-password, verify-code, reset-password, `POST /api/v1/onboarding/schools` and health) and validates it with `GET /api/v1/sessions/{id}` against ms-identity, with a 10 s cache. No `jwt` plugin and no identity headers injected.
- Each service re-validates on its own: session against ms-identity (`/api/v1/sessions/{id}`) and permissions against ms-authorization (`/api/v1/users/{id}/roles` + `/api/v1/auth/evaluate?userId&permission`). Java (01/02/04/05): `AuthTokenFilter`; TypeScript (03/07/08/09): `authGuard.ts`; Python (06): `security.py`.

8.4 Domain Events

Table 22

8.4 Domain Events

Context	Events
Identity	UserRegistered, UserActivated, UserDeactivated, SessionStarted, PasswordChanged
Authorization	RoleAssigned, RoleRemoved, PermissionGranted, PermissionRevoked
Academic	SchoolRegistered, ProgramCreated, CohortRegistered, EnrollmentCreated, CourseAdded, CourseDeleted
Scheduling	EnvironmentRegistered, ScheduleBlockCreated, ScheduleBlockConflictDetected, ClassSessionOpened, ClassSessionClosed

Context	Events
Biometric	FacialEmbeddingEnrolled, FacialEmbeddingUpdated, FingerprintEmbeddingEnrolled, FingerprintEmbeddingUpdated, BiometricUpdateRequested, BiometricUpdateApproved, BiometricUpdateRejected
Attendance	AttendanceRecorded, LateArrivalDetected, AbsenceDetected, JustificationRequested, SupportingDocumentUploaded, JustificationApproved, JustificationRejected
Notification	AlertRaised, AlertResolved

Source. Own elaboration.

Delivery guarantees: At-least-once with idempotent handlers, Dead Letter Queue (DLQ) with 3-5 retries, exponential backoff, 7-day retention.

8.5 OpenAPI Contracts

All services have OpenAPI 3.0.3 specifications in `07-api/contracts/openapi/`:

- `01-identity-service.yaml`
- `02-authorization-service.yaml`
- `03-academic-service.yaml`
- `04-scheduling-service.yaml`
- `05-attendance-service.yaml`
- `06-biometric-service.yaml`
- `07-configuration-service.yaml`
- `08-notification-service.yaml`
- `09-quality-service.yaml`
- `api-gateway.yaml`

9. SECURITY

9.1 Authentication and Authorization

Table 23

9.1 Authentication and Authorization

Mechanism	Details
Authentication	Opaque SessionId (UUID) - never JWT

Mechanism	Details
Password Hashing	BCrypt with cost factor 12, automatic 16-byte salt
Authorization	RBAC with roles: Administrator, Instructor, Learner
Permission evaluation	`GET /api/v1/auth/evaluate?userId&permission` → `{allowed}`
Encryption	SSL/TLS for all communications
Rate Limiting	Kong with Redis (origins: localhost:3000, 5173, 8090)

Source. Own elaboration.

9.2 Data Protection

- *Biometric data as PII:* Facial embeddings and fingerprint templates are treated as sensitive personal information.
- *Law 1581:* Compliance with Colombian personal data protection regulations.
- *Consent:* Explicit consent required for biometric registration.
- *Encryption at rest:* Sensitive data encrypted in the database.
- *Encryption in transit:* All communications use HTTPS/TLS.

9.3 Biometric Validation

- *Liveness Detection (facial):* Liveness verification through blink and movement detection.
- *Fingerprint capture:* Via DigitalPersona 4500 SDK with fingerprint quality validation.
- *Biometric Rate Limiting:* Maximum 20 attempts per 60-second window.
- *Secure storage:* Facial embeddings and fingerprint templates encrypted in MongoDB.

9.4 Security Technical Debt

Table 24

9.4 Security Technical Debt

ID	Description	Priority
AT-004	~~Bearer not validated in TS/Python services (03/06/07/08/09)~~ Resolved: `authGuard.ts` and `security.py` validate session and permissions	—

ID	Description	Priority
AT-005	~~Password recovery endpoints not implemented~~ Resolved: `forgot-password`, `verify-code` and `reset-password` in ms-identity	—
AT-007	Kong does not use a JWT plugin: it validates the opaque session with a *pre-function* (current decision; a signed token is future evolution, ADR-007)	P2

Source. Own elaboration.

10. INFRASTRUCTURE AND DEVOPS

10.1 Environments

Table 25

10.1 Environments

Environment	Description
Development	Local Docker Compose with all dependencies
Staging	Production replica with test data
Production	Deployment with monitoring and alerts

Source. Own elaboration.

10.2 Ports

Table 26

10.2 Ports

Port	Service	Bind
8000	Kong Gateway (proxy)	0.0.0.0
8090	Web Frontend	0.0.0.0
5432	PostgreSQL	127.0.0.1
27017	MongoDB	127.0.0.1
6379	Redis	127.0.0.1
9092	Kafka	127.0.0.1
8081-8089	Microservices	127.0.0.1

Source. Own elaboration.

10.3 Main Environment Variables

```
# PostgreSQL
POSTGRES_USER=postgres
POSTGRES_PASSWORD=YOUR_VALUE
POSTGRES_DB=faceattend_db
POSTGRES_PORT=5432

# MongoDB
MONGO_USER=mongoadmin
MONGO_PASSWORD=YOUR_VALUE
MONGO_DB=faceattend_biometric
MONGO_PORT=27017

# Redis
REDIS_PORT=6379

# Kafka
KAFKA_PORT=9092

# Kong
KONG_PROXY_PORT=8000
KONG_ADMIN_PORT=8001

# Frontend
FRONTEND_PORT=8090
EXPO_PUBLIC_API_URL=

# Biometrics
BIOMETRIC_FACIAL_MATCH_THRESHOLD=0.6
BIOMETRIC_FINGERPRINT_MATCH_THRESHOLD=8
BIOMETRIC_LIVENESS_TTL_SECONDS=120
BIOMETRIC_RATE_LIMIT_MAX_ATTEMPTS=20
BIOMETRIC_RATE_LIMIT_WINDOW_SECONDS=60
BIOMETRIC_LIVENESS_CHALLENGE_SECRET=YOUR_VALUE

# SMTP
SMTP_HOST=host.docker.internal
SMTP_PORT=1025
```

10.4 CI/CD

- *Jenkins* for continuous integration
- *GitHub Actions* for automated checks
- *Liquibase migrations* for database versioning
- *Docker* for consistent packaging

10.5 Observability

- *Logs:* Structured JSON (Pino for TS, structlog for Python)
- *Metrics:* RED (Rate, Errors, Duration)

- *Tracing:* OpenTelemetry for distributed tracing
- *Alerts:* Configured for response time < 5 min

11. WORK PLAN

11.1 Methodology

- *Framework:* Agile with 2-week sprints
- *Approach:* TDD (Test-Driven Development)
- *Documentation:* SDD (Software Design Documentation) - design before code

11.2 Project Phases

Table 27

11.2 Project Phases

Phase	Activities	Duration
Phase 1: Discovery	Problem analysis, user research, benchmarking	Week 1
Phase 2: Definition	Requirements definition, user stories, scope	Week 2
Phase 3: Detailed Design	Architecture, data model, API contracts, ADRs	Weeks 2-3
Phase 4: Implementation	TDD development of microservices and frontend	Weeks 3-10
Phase 5: Testing	Unit, integration, E2E tests, IEEE 829	Weeks 8-12
Phase 6: Deployment	Production configuration, monitoring, go-live	Weeks 12-14

Source. Own elaboration.

11.3 Gantt Chart

Table 28

11.3 Gantt Chart

Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Discovery and Analysis	■■■■					
Design and Architecture	■■■■	■■■■				
Backend Development		■■■■	■■■■	■■■■		
Frontend Development		■■■■	■■■■	■■■■		
Integration				■■■■	■■■■	
Testing and QA				■■■■	■■■■	■■■■

Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Documentation	■■■	■■■	■■■	■■■	■■■	■■■
Deployment						■■■

Source. Own elaboration.

11.4 Deliverables per Sprint

Table 29

11.4 Deliverables per Sprint

Sprint	Deliverables
Sprint 1	Identity service, Authorization service, base data model
Sprint 2	Academic service, Scheduling service, API Gateway
Sprint 3	Attendance service, Biometric service (facial + fingerprint enroll)
Sprint 4	Web Frontend (Login, Dashboard, user management)
Sprint 5	Web Frontend (Attendance, reports), Biometric (identify/verify)
Sprint 6	Justifications, Notifications, Mobile Frontend
Sprint 7	Complete integration, E2E tests, documentation
Sprint 8	Final adjustments, deployment, training

Source. Own elaboration.

12. WORK TEAM

12.1 Members

Table 30

12.1 Members

Name	Role	Category	Responsibility	Contact
Diego Andrés Gutiérrez Nuñez	Administrator, Developer, AI	Software Analysis and Development Technologist Apprentice	Information analysis, design and programming	—
Juan David Arboleda Perdomo	Analyst, Database	Software Analysis and Development Technologist Apprentice	Information analysis, design and programming	—
Jonattan Steven Rizo Solano	Hardware, Developer	Software Analysis and Development Technologist Apprentice	Information analysis, design and programming	—

Source. Own elaboration.

12.2 Instructor

Table 31

12.2 Instructor

Name	Role
Motta Vargas José de Jesús	SENA Instructor

Source. Own elaboration.

13. BUDGET

13.1 Human Resources

Table 32

13.1 Human Resources

Role	Monthly Cost (COP)	Time (months)	Total (COP)
Project Administrator	8,000,000	6	48,000,000
Software Developer (x2)	4,000,000	6	48,000,000
AI/Facial Recognition Specialist	5,000,000	3	15,000,000
Database Administrator	10,000,000	3	30,000,000
Hardware/IoT Engineer	10,000,000	2	20,000,000
TOTAL HUMAN RESOURCES			161,000,000

Source. Own elaboration.

13.2 Vendors and Technologies

Software

Table 33

Software

Vendor	Detail	Cost (COP)
OpenCV / face_recognition	Open Source Software	Free
Amazon Rekognition (alternative)	~4,000 COP per 1,000 images	40,000/month

Vendor	Detail	Cost (COP)
Microsoft Azure Face API (alternative)	~4,000 COP per 1,000 images	40,000/month

Source. Own elaboration.

Hardware

Table 34

Hardware

Vendor	Detail	Cost (COP)
Logitech	1080p cameras (x2)	800,000
Axis Communications	IP cameras (x2)	2,400,000
Dell/HP	Basic server	6,000,000
Raspberry Pi	IoT devices (x2)	400,000

Source. Own elaboration.

13.3 Budget Summary

Table 35

13.3 Budget Summary

Item	Amount (COP)
Human Resources	161,000,000
Hardware (economic option)	7,200,000
Software (Open Source)	0
ESTIMATED TOTAL	168,200,000

Source. Own elaboration.

Estimated minimum budget: $163,800,000 COP

14. QUALITY MANAGEMENT

14.1 Applied Standards

Table 36

14.1 Applied Standards

Standard	Application
IEEE 829	Software test documentation
ISO 25010	Software product quality model
ISO 29110	Processes for small software systems
ISTQB	Software testing certification

Source. Own elaboration.

14.2 Testing Strategy

Table 37

14.2 Testing Strategy

Test Type	Target Coverage	Tool
Unit	≥ 70% (≥ 80% domain)	JUnit, Jest, pytest
Integration	Critical endpoints	Supertest, TestContainers
E2E	Main flows	Detox, Manual
Performance	P95 < 2000ms	k6, JMeter
Security	OWASP Top 10	Manual + tools

Source. Own elaboration.

14.3 Non-Functional Requirements

Table 38

14.3 Non-Functional Requirements

ID	Category	Metric
NFR-001	Performance	P95 < 2000ms critical endpoints, P99 < 5000ms
NFR-002	Availability	Health checks GET /health and /health/ready
NFR-003	Scalability	Support 50% user increase in < 1 year
NFR-004	Security	Opaque sessionId authentication, bcrypt cost 12, RBAC, HTTPS
NFR-005	Compatibility	Chrome, Firefox, Edge, Safari, Brave, Opera
NFR-006	Usability	Registration < 1 min, success rate ≥ 90%, WCAG 2.1 AA
NFR-007	Observability	JSON logs, RED metrics, distributed tracing
NFR-008	Maintainability	Coverage ≥ 70%, cyclomatic complexity ≤ 10

ID	Category	Metric
NFR-009	Portability	Docker images, environment variables
NFR-010	Recovery	RTO < 5 min, RPO 0, daily backups
NFR-011	Integrity	Unique record, immutable evidence, audit

Source. Own elaboration.

15. RISK MANAGEMENT

15.1 Identified Risks

Table 39

15.1 Identified Risks

ID	Risk	Probability	Impact	Mitigation
R-01	Facial recognition failure under variable conditions	High	High	Liveness detection, multiple attempts, manual fallback
R-02	User resistance to change	Medium	High	Training, intuitive interface, continuous support
R-03	Connectivity issues in the classroom	Medium	High	Offline mode (future), manual backup registration
R-04	Project scope growth	High	Medium	Clear MVP definition, roadmap by horizons
R-05	Data regulation non-compliance	Low	High	Legal audit, encryption, explicit consent
R-06	Accumulated technical debt	Medium	Medium	Continuous refactoring, code reviews
R-07	Dependency on a single PostgreSQL	Medium	High	Migration plan to separate instances

Source. Own elaboration.

15.2 Registered Technical Debt

Table 40

15.2 Registered Technical Debt

ID	Description	Priority
AT-001	Single PostgreSQL instance is SPOF	P1

ID	Description	Priority
AT-002	No import-boundary lint rule	P2
AT-003	~~Go runtime adds 4th runtime for 2 tables~~ Resolved: ms-notification was rewritten in TypeScript	—
AT-004	~~Bearer not validated in TS/Python services~~ Resolved	—
AT-005	~~Password recovery endpoints not implemented~~ Resolved	—
AT-006	No circuit breaker, transactional outbox, or Kafka DLQ	P1
AT-007	Kong does not use a JWT plugin: it validates the opaque session with a *pre-function* (see ADR-007)	P2

Source. Own elaboration.

16. ANNEXES

16.1 Glossary of Terms

Table 41

16.1 Glossary of Terms

Term	Definition
Bounded Context	Delimited context in DDD where a domain model is consistent
Liveness Detection	Verification that the captured face belongs to a living person
Embedding	128-dimensional numerical vector representing facial characteristics
SessionId	Opaque identifier (UUID) of an `identity.user_session` row; used as the Bearer token and validated against ms-identity
RBAC	Role-based access control
Saga	Pattern for managing distributed transactions
DLQ	Dead Letter Queue - queue of messages that could not be processed
ADR	Architecture Decision Record - architectural decision document
TDD	Test-Driven Development - test-driven development
SDD	Software Design Documentation - software design documentation

Source. Own elaboration.

16.2 References

Congress of Colombia. (2012). *Statutory Law 1581 of 2012, which establishes general provisions for the protection of personal data* (Diario Oficial No. 48.587).

Institute of Electrical and Electronics Engineers. (2008). *IEEE standard for software and system test documentation* (IEEE Std 829-2008).

International Organization for Standardization & International Electrotechnical Commission. (2011). *Systems and software engineering — Systems and software quality requirements and evaluation (SQuaRE) — System and software quality models* (ISO/IEC 25010:2011). https://www.iso.org/standard/35733.html

International Organization for Standardization & International Electrotechnical Commission. (2016). *Systems and software engineering — Lifecycle profiles for very small entities (VSEs) — Part 1: Overview* (ISO/IEC TR 29110-1:2016).

International Software Testing Qualifications Board. (2018). *Certified Tester Foundation Level syllabus*. ISTQB.

16.3 Repository

- *Documentation:* `fae-docs/`
- *Source code:* `ProyectoFaceAttendEDU/`

Activity plan

Roles (Scrum)

Table 42

Roles (Scrum)

Scrum Role	Project Role	Member	Main Responsibility
Product Owner (PO)	Technical Instructors	Technical Instructors	In charge of reviewing, validating, and approving product progress at each Sprint closing.
Scrum Master (SM)	Developer	Diego Andres Gutierrez Nuñez	Facilitates the Scrum framework, supports the team to follow proper practices, and removes obstacles affecting development.
Developer	Analyst	Juan David Arboleda Perdomo	Actively participates in the creation of the Product Increment during each Sprint.
Developer	Developer Analyst	Jonathan Steven Rizo Solano	Collaborates in the construction of planned functionalities for each Sprint.

Source. Own elaboration.

Activity Plan

Table 43

Activity Plan

No.	Activities	Objectives	Participants	Responsible	Schedule
1	Information collection and analysis	Ask what needs are identified when taking attendance at a conglomerate institution	All members	Developer	1 week of the first stipulated month
1	Information collection and analysis	Analyze collected information and propose a solution that satisfies the identified problem	All members	Analyst	2 week of the first stipulated month
2	Application planning and structuring	Define and agree on application bases (flows, initial cases, possible cases, etc.)	All members	Analyst and Developer	3 and 4 week of the first stipulated month
2	Application planning and structuring	Analyze and search for exceptions in the initially identified flow	Analyst and Developer Analyst	Analyst and Developer Analyst	1 and 2 week of the second stipulated month

No.	Activities	Objectives	Participants	Responsible	Schedule
3	Initial application development	Analyze and agree on the database to be used by the project	All members	Developer	3 and 4 week of the second stipulated month
3	Initial application development	Development and start of database testing phase	All members	Developer and Analyst	1 week of the third stipulated month
3	Initial application development	Start of application logic development	All members	Analyst and Developer Analyst	2, 3 and 4 week of the third stipulated month
3	Initial application development	Application control testing in initial stage	Analyst and Developer	Analyst	4 week of the fourth stipulated month
4	Mid application development	Start of business logic development	All members	Developer	1 and 2 week of the fourth stipulated month
4	Mid application development	Start of web interface development	Analyst and Developer	Developer Analyst	2, 3 and 4 week of the fourth stipulated month
4	Mid application development	Start of mobile interface development	Analyst Developer and Developer	Developer Analyst	2, 3 and 4 week of the fourth stipulated month
4	Mid application development	Control testing of database implementation and application interfaces	Analyst and Developer	Analyst	1 and 2 week of the fifth stipulated month
5	Preliminary application development	Correction and control of errors and exceptions found during each control test	All members	Analyst and Developer Analyst	3 and 4 week of the fifth stipulated month
5	Preliminary application development	Improvement of application details and features	All members	Developer and Analyst	4 week of the fifth stipulated month
6	Final application development and delivery	Control testing of the final application	All members	All members	1, 2, 3 and 4 week of the sixth stipulated month

Source. Own elaboration.

Hardware requirements

Sheet: Developers

Proposal 1 (initial)

Project name: FACEATTEND EDU *Members:* Diego Andres Gutierrez Nuñez, Jonathan Stiven Rizo, Juan David Arboleda Perdomo

Table 44

Proposal 1 (initial)

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Laptop	Intel Core I3 10th generation; Kingston 16GB RAM 4400MT/s DDR4; Adata 120GB SSD; 14.5-inch screen	Programming	This device is required to code the project, these specifications are the most optimized for project management and coding	3	$1,689,900	$5,069,700	https://www.falabella.com.co/falabella-co/product/140970655/PORTATIL-DELL-INTEL-CORE-I3-1215U-1TB-SSD-16GB-RAM-156-FHD/140970656?	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Desktop PC	Intel Core I5 7th generation; Kingston 16GB RAM 550MT/s DDR4; Adata 240GB SSD; 24-inch Monitor	Tester	This device is required to test the project, these specifications are the most optimized for testing and error/bug verification	1	$1,119,027	$1,119,027	https://encarguelo.com/producto/B0BN1VRT4T/dell-computadora-de-escritorio-optiplex-7050-intel-core-i5-7500-340ghz-16gb-ddr4-ram-1tb-ssd-wi-fi-integrado-bluetooth-windows-11-pro-soporte-4k-hd-graphics-630-renovado?	—
Mouse	Generic Logitech optical wired USB mouse	Peripheral	This peripheral is required for control and data transfer from the physical to digital world, to manipulate data recorded on the PC screen	3	$38,250	$114,750	Mouse Logitech Alambrico Optico USB Negro M110 SILENT 910-005493	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Keyboard	Generic Hewlett-Packard mechanical wired USB keyboard	Peripheral	This peripheral is required for data entry into the PC, ability to modify values and better use of the PC	3	$65,150	$195,450	Amazon.com: HP Hewlett Packard Business Negro USB Slim Style Windows Teclado mejorado P/N 803823-001 : Electrónica	—
Server	Intel Core I5 7th generation; Kingston 24GB RAM 550MT/s DDR4; Adata 512GB SSD; 24-inch Monitor	System Hosting	This device is required for project use and execution, these specifications are the most optimized for efficient program use and performance	1	$1,099,999	$1,099,999	Torre Dell Optiplex 7050 Micro Core I5 7th 8gb Ssd 240gb	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Modem	Dual Band 2.4GHz - 5GHz; WiFi 5 / WiFi 6; Download Speed 30 Mbps - Upload Speed 20 Mbps	IoT Connectivity	This device is required for a stable connection between the server and the cloud to allow correct system registration	1	$179,900	$179,900	https://www.alkosto.com/router-tp-link-wifi-6-4-antenas-ax1500-mbps-archer-ax12/p/8400307053 80?	—
Optical Camera	2K Optical Camera; CMOS sensor; 30-40 fps; FOV 70°-90°	Image Capture	This peripheral is required for capturing the user's optical image to initiate the registration process through the application and cloud registration	1	$96,600	$96,600	https://www.mercadolibre.com.co/camara-web-usb-full-hd-2k-para-computadora-con-microfono/p/MCO2 00668708 7? matt_tool =1939012 7&pdp_fil ters=item _id%3A MCO233 0707766 &from=gs hop	—
Total						$7,875,42 6		

Source. Own elaboration.

Proposal 2 (improved)

Project name: FACEATTEND EDU *Members:* Diego Andres Gutierrez Nuñez, Jonathan Stiven Rizo, Juan David Arboleda Perdomo

Table 45

Proposal 2 (improved)

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Laptop	Intel Core i5 11th Gen; 16GB RAM DDR4; SSD 256GB NVMe; 15.6" FHD Screen	Programming	A laptop with better processor and storage than the initial proposal is required, to ensure fluency in compilation and multitasking	3	$2,099,000	$6,297,000	https://www.falabella.com.co/falabella-co/product/140970655/PORTATIL-DELL-INTEL-CORE-I3-1215U-1TB-SSD-16GB-RAM-156-FHD/140970656?	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Desktop PC	Intel Core i7 8th Gen; Intel Core i7 8th Gen; SSD 512GB; 24" Monitor	Tester	Testing requires more graphical capacity and fast storage, ensuring stability in simulations and stress tests	1	$1,699,000	$1,699,000	https://encarguelo.com/producto/B0BN1VRT4T/dell-computadora-de-escritorio-optiplex-7050-intel-core-i5-7500-340ghz-16gb-ddr4-ram-1tb-ssd-wi-fi-integrado-bluetooth-windows-11-pro-soporte-4k-hd-graphics-630-renovado?	—
Mouse	Logitech M185 Wireless	Peripheral	This peripheral is required for control and data transfer from the physical to digital world, to manipulate data recorded on the PC screen	3	$54,900	$164,700	https://www.alkosto.com/mouse-logitech-inalambrico-m185/p/097855067122	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Keyboard	Logitech K120 USB	Peripheral	This peripheral is required for data entry into the PC, ability to modify values and better use of the PC	3	$49,900	$149,700	https://www.mercadolibre.com.co/teclado-logitech-k120-usb/p/MCO6052210	—
Server	Intel Core i7 9th Gen; 32GB RAM DDR4; SSD NVMe 1TB	System Hosting	This device is required for project use and execution, these specifications are the most optimized for efficient program use and performance	1	$3,200,000	$3,200,000	Torre Dell Optiplex 7050 Micro Core I5 7th 8gb Ssd 240gb	—
Modem	Router TP-Link WiFi 6 AX1800	IoT Connectivity	This device is required for a stable connection between the server and the cloud to allow correct system registration	1	$249,900	$249,900	https://www.alkosto.com/router-tp-link-archer-ax20-ax1800/p/840030704605	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Optical Camera	Logitech C920s Pro; Full HD 1080p; 30 fps; FOV 78°	Image Capture	This peripheral is required for capturing the user's optical image to initiate the registration process through the application and cloud registration	1	$279,900	$279,900	https://www.mercadolibre.com.co/camara-web-usb-full-hd-2k-para-computadora-con-microfono/p/MCO200668708 7?matt_tool=19390127&pdp_filters=item_id%3AMCO2330707766&from=gshop	—
Total						$12,040,200		

Source. Own elaboration.

Sheet: Software

Note: the content of this sheet is identical to Proposal 1 from the Developers sheet.

Project name: FACEATTEND EDU *Members:* Diego Andres Gutierrez Nuñez, Jonathan Stiven Rizo, Juan David Arboleda Perdomo

Table 46

Sheet: Software

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Laptop	Intel Core I3 10th generation; Kingston 16GB RAM 4400MT/s DDR4; Adata 120GB SSD; 14.5-inch screen	Programming	This device is required to code the project, these specifications are the most optimized for project management and coding	3	$1,689,900	$5,069,700	https://www.falabella.com.co/falabella-co/product/140970655/PORTATIL-DELL-INTEL-CORE-I3-1215U-1TB-SSD-16GB-RAM-156-FHD/140970656?	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Desktop PC	Intel Core I5 7th generation; Kingston 16GB RAM 550MT/s DDR4; Adata 240GB SSD; 24-inch Monitor	Tester	This device is required to test the project, these specifications are the most optimized for testing and error/bug verification	1	$1,119,027	$1,119,027	https://encarguelo.com/producto/B0BN1VRT4T/dell-computadora-de-escritorio-optiplex-7050-intel-core-i5-7500-340ghz-16gb-ddr4-ram-1tb-ssd-wi-fi-integrado-bluetooth-windows-11-pro-soporte-4k-hd-graphics-630-renovado?	—
Mouse	Generic Logitech optical wired USB mouse	Peripheral	This peripheral is required for control and data transfer from the physical to digital world, to manipulate data recorded on the PC screen	3	$38,250	$114,750	Mouse Logitech Alambrico Optico USB Negro M110 SILENT 910-005493	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Keyboard	Generic Hewlett-Packard mechanical wired USB keyboard	Peripheral	This peripheral is required for data entry into the PC, ability to modify values and better use of the PC	3	$65,150	$195,450	Amazon.com: HP Hewlett Packard Business Negro USB Slim Style Windows Teclado mejorado P/N 803823-001 : Electrónica	—
Server	Intel Core I5 7th generation; Kingston 24GB RAM 550MT/s DDR4; Adata 512GB SSD; 24-inch Monitor	System Hosting	This device is required for project use and execution, these specifications are the most optimized for efficient program use and performance	1	$1,099,999	$1,099,999	Torre Dell Optiplex 7050 Micro Core I5 7th 8gb Ssd 240gb	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Modem	Dual Band 2.4GHz - 5GHz; WiFi 5 / WiFi 6; Download Speed 30 Mbps - Upload Speed 20 Mbps	IoT Connectivity	This device is required for a stable connection between the server and the cloud to allow correct system registration	1	$179,900	$179,900	https://www.alkosto.com/router-tp-link-wifi-6-4-antenas-ax1500-mbps-archer-ax12/p/8400307053 80?	—
Optical Camera	2K Optical Camera; CMOS sensor; 30-40 fps; FOV 70°-90°	Image Capture	This peripheral is required for capturing the user's optical image to initiate the registration process through the application and cloud registration	1	$96,600	$96,600	https://www.mercadolibre.com.co/camara-web-usb-full-hd-2k-para-computadora-con-microfono/p/MCO2 00668708 7? matt_tool =1939012 7&pdp_fil ters=item _id%3A MCO233 0707766 &from=gs hop	—
Total						$7,875,42 6		

Source. Own elaboration.

Sheet: Users

Proposal 1 (initial)

Program name: FACEATTEND EDU *Members:* Diego Andres Gutierrez Nuñez, Jonathan Stiven Rizo, Juan David Arboleda Perdomo

Table 47

Proposal 1 (initial)

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Server	Intel Core I5 7th generation; Kingston 24GB RAM 550MT/s DDR4; Adata 512GB SSD; 24-inch Monitor	System Hosting	This device is required for program use and execution, these specifications are the most optimized for efficient program use and performance	1	$1,099,999	$1,099,999	Torre Dell Optiplex 7050 Micro Core I5 7th 8gb Ssd 240gb	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Mouse	Generic Logitech optical wired USB mouse	Peripheral	This peripheral is required for control and data transfer from the physical to digital world, to manipulate data recorded on the PC screen	3	$38,250	$114,750	Mouse Logitech Alambrico Optico USB Negro M110 SILENT 910-005493	—
Keyboard	Generic Hewlett-Packard mechanical wired USB keyboard	Peripheral	This peripheral is required for data entry into the PC, ability to modify values and better use of the PC	3	$65,150	$195,450	Amazon.com: HP Hewlett Packard Business Negro USB Slim Style Windows Teclado mejorado P/N 803823-001 : Electrónica	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Modem	Dual Band 2.4GHz - 5GHz; WiFi 5 / WiFi 6; Download Speed 30 Mbps - Upload Speed 20 Mbps	IoT Connectivity	This device is required for a stable connection between the server and the cloud to allow correct system registration	1	$179,900	$179,900	https://www.alkosto.com/router-tp-link-wifi-6-4-antenas-ax1500-mbps-archer-ax12/p/8400307053 80?	—
Optical Camera	2K Optical Camera; CMOS sensor; 30-40 fps; FOV 70°-90°	Image Capture	This peripheral is required for capturing the user's optical image to initiate the registration process through the application and cloud registration	1	$96,600	$96,600	https://www.mercadolibre.com.co/camara-web-usb-full-hd-2k-para-computadora-con-microfono/p/MCO2 00668708 7? matt_tool =1939012 7&pdp_fil ters=item _id%3A MCO233 0707766 &from=gs hop	—
Total						$1,686,69 9		

Source. Own elaboration.

Proposal 2 (reformulated)

Program name: FACEATTEND EDU *Members:* Diego Andres Gutierrez Nuñez, Jonathan Stiven Rizo, Juan David Arboleda Perdomo

Table 48

Proposal 2 (reformulated)

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Server	Intel Core i5 7th generation; Kingston 24GB RAM DDR4 550MT/s; Adata 512GB SSD; 24" Monitor	System Hosting	This device is required for program use and execution, these specifications are the most optimized for efficient program use and performance	1	$1,099,999	$1,099,999	Torre Dell Optiplex 7050 Micro Core I5 7th 8gb Ssd 240gb	—
Mouse	Generic Logitech optical wired USB	Peripheral	This peripheral is required for control and data transfer from the physical to digital world, to manipulate data recorded on the PC screen	3	$38,250	$114,750	Mouse Logitech Alambrico Optico USB Negro M110 SILENT 910-005493	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Keyboard	Hewlett-Packard mechanical USB	Peripheral	This peripheral is required for data entry into the PC, ability to modify values and better use of the PC	3	$65,150	$195,450	https://www.mercadolibre.com.co/camara-web-usb-full-hd-2k-para-computadora-con-microfono/p/MCO200668708 7?matt_tool=19390127&pdp_filters=item_id%3AMCO2330707766&from=gshop	—
Modem	Dual Band 2.4GHz – 5GHz; WiFi 5 / WiFi 6; Speed 30 Mbps Down / 20 Mbps Up	IoT Connectivity	This device is required for a stable connection between the server and the cloud to allow correct system registration	1	$179,900	$179,900	https://www.alkosto.com/router-tp-link-wifi-6-4-antenas-ax1500-mbps-archer-ax12/p/8400307053 80?	—

Requirement	Specifications	Use	Justification	Quantity	Unit Value (COP)	Total Value (COP)	Supplier	Approver
Optical Camera	2K Resolution; CMOS sensor; 30-40 fps; FOV 70°–90°	Image Capture	This peripheral is required for capturing the user's optical image to initiate the registration process through the application and cloud registration	1	$96,600	$96,600	https://www.mercadolibre.com.co/camara-web-usb-full-hd-2k-para-computadora-con-microfono/p/MCO200668708 7?matt_tool=19390127&pdp_filters=item_id%3AMCO2330707766&from=gshop	—
Total						$1,686,699		

Source. Own elaboration.